Justify all answers. Submit as a single PDF.

1. Let $C \subseteq \{0,1\}^4$ be the code $C = \{0000,\ 0101,\ 1010,\ 1111\}$.
   (a) (5 points) Determine the minimum distance $d(C)$.
   (b) (5 points) What is the largest number of bit errors that this code can guarantee to detect? What is the largest number of bit errors that it can guarantee to correct?

2. Let C be the binary code $C = \{(0,0,1),\ (1,1,1),\ (1,0,0),\ (0,1,0)\}$.
   (a) (5 points) Show that C is not linear.
   (b) (5 points) What is $d(C)$? How many errors can this code detect and correct?

3. Consider transmission over a binary symmetric channel with crossover (error) probability $p < 1/2$. Let $C \subseteq \{0,1\}^5$ be the code $C = \{00000,\ 11100,\ 00111,\ 11011\}$. Suppose the received word is $y = 00100$.
   (a) (5 points) For every codeword $x \in C$, compute $\Pr(y \mid x \text{ sent})$.
   (b) (5 points) Using maximum likelihood decoding, which codeword in C should be chosen as the estimate of the transmitted word?

4. Work over the binary field $\mathbb{F}_2$. Consider the generator matrix

$$G = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \end{bmatrix}.$$

   Let $C = \{mG : m \in \mathbb{F}_2^2\}$.
   (a) (5 points) How many codewords does C have?
   (b) (5 points) List all codewords in C explicitly.
   (c) (5 points) Compute the minimum distance $d(C)$. How many errors can this code detect and correct?
   (d) (5 points) Find a parity-check matrix H such that $C = \{c \in \mathbb{F}_2^5 : Hc^\top = 0\}$.

5. The following is a parity-check matrix for a binary $[n,k]$ code C:

$$H = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix}.$$

   (a) (5 points) Find n and k.
   (b) (5 points) Find the generator matrix for C.
   (c) (5 points) List the codewords in C.
   (d) (5 points) What is the code rate for C?

6. (20 points) Consider the $(7,4)$ Hamming code with generator and parity-check matrices

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}, \quad H = \begin{bmatrix} 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}.$$

All arithmetic is over $\mathbb{F}_2$. Two codewords were sent using the Hamming $[7,4]$ code and were received as 0100111 and 0101010. Each one contains at most one error. Correct the errors. Also, determine the 4-bit messages that were multiplied by the matrix G to obtain the codewords.

7. Let X be the outcome of a fair coin toss: $\Pr(X = \mathrm{H}) = \Pr(X = \mathrm{T}) = 1/2$. Let Y be a biased coin toss with $\Pr(Y = \mathrm{H}) = 3/4$ and $\Pr(Y = \mathrm{T}) = 1/4$. Let Z be a random variable taking values $\{a, b, c\}$ with probabilities

$$\Pr(Z = a) = \tfrac{1}{2}, \quad \Pr(Z = b) = \tfrac{1}{4}, \quad \Pr(Z = c) = \tfrac{1}{4}.$$

All entropies should be computed in bits (base-2 logarithms).

(a) (5 points) Compute $H(X)$, $H(Y)$, and $H(Z)$.

(b) (5 points) Which of X, Y, Z has the greatest entropy? Which has the least? Explain how this matches your intuition about how "uncertain" each random variable is.

8. Consider a random variable X taking values in the alphabet $\mathcal{A} = \{A, B, C, D, E\}$ with probabilities

$$\Pr(A) = 0.40, \quad \Pr(B) = 0.20, \quad \Pr(C) = 0.20, \quad \Pr(D) = 0.10, \quad \Pr(E) = 0.10.$$

(a) (5 points) Compute the Shannon entropy $H(X)$ (in bits).

(b) (5 points) Use the Huffman algorithm to construct a binary code for X. (Give the codeword for each symbol.) Compute the average codeword length L for your code and verify that $H(X) \leq L < H(X) + 1$.

9. Consider the alphabet $\{\#, A, B, \ldots, Z\}$ with initial dictionary

$$\# \mapsto 0, \quad A \mapsto 1, \; \ldots, \; Z \mapsto 26.$$

(a) (5 points) Apply the Lempel–Ziv–Welch (LZW) algorithm to the string

`PETERPIPERPICKEDAPECKOFPICKLEDPEPPERS#`.

List the output bits and the new dictionary entries. *Hint: Start with 5-bit codes. Once the dictionary has more than* 32 *entries, you will need 6-bit codes.*

(b) (5 points) Compare the output with naively encoding the message using only the initial alphabet. How many bits did LZW save?

10. (20 points) You are given 16 coins, all of which are equal in weight except for one that is either heavier or lighter. You are also given a bizarre two-pan balance that can report only two outcomes: "the two sides balance" or "the two sides do not balance". Design a strategy to determine which is the odd coin (you do not need to determine whether it is heavier or lighter) in as few uses of the balance as possible. *Hint: Each weighing is a yes/no question of the form "Is the odd coin among this set?".*